

## Glycemic responses to sweetened dried and raw cranberries in humans with type 2 diabetes.



This study assessed the metabolic response to sweetened dried cranberries (SDC), raw cranberries (RC), and white bread (WB) in humans with type 2 diabetes. Development of palatable cranberry preparations associated with lower glycemic responses may be useful for improving fruit consumption and glycemic control among those with diabetes. In this trial, type 2 diabetics (n= 13) received WB (57 g, 160 cal, 1 g fiber), RC (55 g, 21 cal, 1 g fiber), SDC (40 g, 138 cal, 2.1 g fiber), and SDC containing less sugar (SDC-LS, 40 g, 113 cal, 1.8 g fiber + 10 g polydextrose). Plasma glucose (mmol/L) peaked significantly at 60 min for WB, and at 30 min for RC, SDC, and SDC-LS at  $9.6 \pm 0.4$ ,  $7.0 \pm 0.4$ ,  $9.6 \pm 0.5$ , and  $8.7 \pm 0.5$ , respectively, WB remained significantly elevated from the other treatments at 120 min. Plasma insulin (pmol/mL) peaked at 60 min for WB and SDC and at 30 min for RC and SDC-LS at  $157 \pm 15$ ,  $142 \pm 27$ ,  $61 \pm 8$ , and  $97 \pm 11$ , respectively. Plasma insulin for SDC-LS was significantly lower at 60 min than either WB or SDC. Insulin area under the curve (AUC) values for RC and SDC-LS were both significantly lower than WB or SDC. Phenolic content of SDC and SDC-LS was determined following extraction with 80% acetone prior to high-performance liquid chromatography (HPLC) and electrospray ionization-mass spectrometry (ESI-MS) and found to be rich in 5-caffeoylquinic acid, quercetin-3-galactoside, and the proanthocyanidin dimer epicatechin. In conclusion, SDC-LS was associated with a favorable glycemic and insulinemic response in type 2 diabetics. Practical Application: This study compares phenolic content and glycemic responses among different cranberry products. The study seeks to expand the palatable and portable healthy food choices for persons with type 2 diabetes. The novel use of polydextrose as a bulking agent making possible a reduction in caloric content and potential glycemic response is also characterized in this study.